

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (Mechanical) (Credit System) Degree Examinations**

April 2023

21ME302 – MANUFACTURING PROCESSES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Suggest a suitable layout for Assembly of aero plane. List its salient features, advantages and disadvantages.	7	L*1	1	1
	b) Identify the pattern material used for the following application and explain their salient characteristics i) Small and intricate patterns ii) Pattern for investment casting iii) Patterns used in machine molding.	6	L3	1	1
	c) Illustrate the working of a molding machine with a sketch which provides uniform ramming of the sand in all portions of the moulds.	7	L3	1	1
2.	a) The selection of plant layout is dependent on several factors. Elaborate.	3	L2	1	1
	b) Sketch and explain the constructional features and working principles of cupola furnace.	10	L2	1	1
	c) Identify a component which is used to develop cavities in the casting. Explain with a sketch, any three types of it depending on shape and position of the mould.	7	L3	1	1
3.	a) It is observed that in a molding process, sand mold is held together by thermosetting resin binder. With neat sketches explain various steps involved in it and list its advantages and disadvantages.	8	L2	1	1
	b) Exemplify any 3 types of casting defects and list its causes and solutions to eliminate these defects.	6	L3	1	1
	c) Define pattern allowance. Explain any 5 types of it.	6	L3	1	1
Unit – II					
4.	a) What do you mean by rolling? Explain with sketch, planetary rolling mill and tandem rolling mill in detail.	7	L2	2	1
	b) How do you manufacture hollow storage tanks made of plastic? Explain with a neat sketch.	7	L3	2	1
	c) With examples explain the various ceramic materials used in the manufacturing industries.	6	L2	2	1
5.	a) With a neat sketch explain the process used to manufacture snack tubs made of plastics.	7	L2	2	1
	b) Critically evaluate the techniques used to process ceramics. Briefly explain with a block diagram.	8	L1	2	1
	c) With sketch explain the working principle of compound dies.	5	L1	2	1
6.	a) With a sketch explain a process used to manufacture toothpaste tubes.	6	L2	2	1

P.T.O.

- b) Sketch and explain following Piece Ware Shaping Processes.

- i. Spinning,
- ii. Sagging,
- iii. Blowing,
- iv. Pressing and blow forming

8 L2 2 1

- c) With sketch, compare two types of atomization process used in powder metallurgy.

6 L2 2 1

Unit – III

7. a) It is required to join railway lines. Explain a suitable welding process used to it with relevant sketches.

7 L3 3 1

- b) Write a note on welding defects.

7 L2 3 1

- c) Critically evaluate the adhesive bonding process and list its merits and demerits.

6 L2 3 1

8. a) Describe with a sketch, an arc welding process, which uses a non-consumable electrode.

6 L3 3 1

- b) With a block diagram, explain the principle of resistance welding. List some of the de merits of it.

6 L2 3 1

- c) Assess the Electron beam welding process and list its advantages and limitations.

8 L1 3 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
